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**Textometric analysis of Swiss spatial
planning policies**

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Abstract

This study investigates how accessibility, motility, and time are represented in Swiss spatial planning discourse through a textometric approach.

It does so using a corpus made of 74 plans accross all cantons of switzerland spanning between 1987 to 2025. The results reveal a pronounced imbalance: accessibility dominates and is expressed in concrete, operational terms, while the other aspect of motility, skills, motivation, and cognitive appropriation is almost entirely absent. This absence shows that capability-based understandings of mobility remain outside the institutional vocabulary.

Time-related notions display a similar asymmetry. Objective and measurable aspects of time are frequently mentioned, whereas perceived or subjective time appears only marginally. Concepts such as comfort are present but vaguely defined, suggesting that experiential dimensions of mobility are acknowledged but not yet integrated into planning practice.

The textometric method proved effective in exposing what institutional discourse emphasizes and what it leaves unspoken. Furthermore, this approach, while not perfect, reduce researcher bias by offering some quantitative result.

The project contributes both by offering an analysis that show a discursive bias toward technical framings of mobility and an open, dataset with a CC-BY license to support further research.

Overall, the findings underline the need for conceptual and institutional shifts to incorporate motility and experiential time into planning. Doing so would help move Swiss mobility policy toward mobility policies that better account for citizen capabilities and experiential aspects of travel, and thus help transition to a more sustainable mobility.

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1 Introduction

1.1 Background: Spatial Planning and Transport Policies in Switzerland

Spatial planning in Switzerland operates in a layered fashion. The federal Spatial Planning Act (LAT: Loi sur l'aménagement du territoire) [2] was introduced in 1979 by the Confederation and apply to each Cantons and Communes. It defines the overall framework and general objectives such as containing urban sprawl, protecting natural resources, and ensuring efficient land use, but delegates the implementation to the Cantons. Each Canton is responsible for elaborating a "cantonal master plan" (plan directeur cantonal) within the first eight years of the LAT's introduction and then revising it at least every ten years. The Confederation must review and approve each plan and its revisions.

These plans translate federal principles into territorial strategies that account for local geography and specific needs. Once approved by the Confederation, they become binding for public authorities but remain indicative for private actors, shaping how subsequent zoning and land-use decisions are made. The Cantons delegate the responsibility for implementing some of the measures to the Communes, with the main exception of cantonal infrastructure such as cantonal roads, which remain under cantonal responsibility.

The Communes achieve the goals either by adopting local regulations (règlement communal) or by directly implementing measures, for instance by creating new bike lanes. Some Communes also prepare a "local master plan" (plan d'affectation communal), which materializes these strategies by determining the precise allocation of land and the regulations governing its use.

This nested structure offer flexibility to adapt the general goal as close as possible to the local reality but it require a fair amount of effort to ensure the efficient communication between all the actors. Over time, revisions of the LAT, most notably in 2014 [7], shifted the discourse from expansion toward densification, sustainability, and the coordination of land use with transport systems. The vast majority of early plans exist only as paper archives available only trough physical consultation, limiting historical analysis, but comparative studies between cantonal documents remain meaningful. Each plan reveals how spatial and transport policies adapt a shared legal foundation to diverse territorial realities and political cultures.

1.2 Research Motivation: Linking Travel Behavior, Planning, and Text Analysis

Effective policy design requires a systemic perspective. Interventions that optimize without taking into account the system as a whole often if not always generate unintended consequences [14]. The problem is actually very similar to what arise in artificial intelligence when systems end up acting in unanticipated ways due to failure (among other factor) of the cost function to capture the entirety of the human intent with the only goal of minimize the said cost function [13].

This principle directly applies to transportation policy. One of its main goals is to reduce environmental harm by lowering greenhouse gas emissions, which are the primary cause of human-induced climate change. The transport sector plays a key role in this effort. In Europe, it is the only major sector where emissions have not decreased, and it remains the second-largest emitter after energy production [3]. Decarbonizing transport is therefore essential for mitigating climate change. Furthermore, greenhouse gases are not the only negative externality. Transport also generates tire-related microplastics, fine particles, noise pollution, accident risks, soil sealing, and barriers to animal movement. These impacts must be balanced with other objectives derived from the right to mobility [17], such as equity and inclusivity, economic efficiency, and safety. Ultimately, transport policy must navigate a tradeoff between protecting the environment and its inhabitants from harmful externalities and ensuring the human need for mobility. This is not a trivial task as both factor are complex with many interconnected variables.

A transition to active mobility and public transport offers the most equitable path forward. Private vehicles are highly inefficient in their use of urban space. They provide primary benefits to a minority of users. This often occurs at a significant cost to the broader community [12]. Prioritizing sustainable modes is a matter of fairness and justice [9].

However, the existence of sustainable infrastructure does not guarantee its use. Research, such as the work of J. Grandvillemin and Al (not published yet), demonstrates this clearly. Modal choice is not a purely rational decision. It is shaped by intangible factors like personal habits, preferences, and perceived access. The sociological concept of motility effectively captures this capacity for mobility [11]. Understanding these human factors is crucial for designing effective systems.

This leads to the core research question of this document. To what degree are these human dimensions of mobility integrated into Swiss spatial planning? This study investigates whether cantonal master plans consider the factors that truly shape travel behavior. It aims to determine if planning discourse addresses how people experience time and choose between mobility options. The ultimate goal is to assess if Swiss policy aligns physical infrastructure with the human elements of mobility and thus to offer a path to improve the future versions of said plans.

1.3 Research Questions

To what extent do Swiss territorial / spatial planning documents acknowledge the distinction between perceived time vs measured/travel-time objectively modeled, in relation to modal choice?

How strongly do Swiss planning documents incorporate human / behavioral / motility-based rationales, as opposed to purely economic / efficiency rationales, in shaping transport and spatial policy?

2 State of the Art and Conceptual Framework

This research draws on the sociology of mobility, a field that reorients analysis away from the mere observation of movement and toward the conditions, capacities, and lived experiences that shape how individuals actually move through space. Before reviewing the relevant literature, two foundational concepts must be clarified: *motility* and *accessibility*. Common-sense understandings of mobility tend to collapse these distinctions, equating mobility with the infrastructural provision of access. This reduction obscures the fact that mobility emerges not only from the physical availability of transport options but also from the skills, perceptions, habits, and motivations that condition whether and how individuals make use of them. Motility encompasses these multiple dimensions and thus provides a richer analytical framework for examining the fracture that frequently appear between planned transport offerings and actual mobility practices.

2.1 State of the Art

Historically, mobility has been conceptualized within an engineering-based paradigm that evaluates systems primarily through quantifiable performance metrics. Geurs and van Wee’s foundational review of accessibility measures [8] maps this landscape in detail. They distinguish between various modeling traditions and highlight their analytical strengths and limitations. While these refinements offer increasing sophistication in representing spatial opportunities, even the most advanced frameworks remain grounded in a functionalist logic: accessibility is understood as an objective property of the system, measurable through indicators such as travel time, cost, and network reachability.

The central tension identified by Geurs and van Wee is that operational planning often relies on simplified indicators (e.g., average speed, congestion levels) meant to ease communication and decision-making. This simplification tends to obscure the behavioral and experiential dimensions of travel. Although the authors call for the integration of feedback mechanisms and spatial-behavioral interactions, their framework ultimately still treats accessibility as a supply-side phenomenon. What remains largely absent is any account of how individuals make sense of, value, or emotionally respond to the mobility options that planners design for them.

It is precisely this gap that sociological approaches to mobility seek to fill. Sheller and Urry’s “new mobilities paradigm” [20] reposition mobility as a socio-material assemblage shaped by technologies, infrastructures, cultural meaning, and embodied practices. Within this broader paradigm, Kaufmann’s concept of motility [11] has become a key theoretical tool for understanding the potential for movement, rather than movement itself. Motility decomposes this potential into three interacting components: access, skills, and cognitive appropriation.

Doherty’s notion of *viscosity* [6] adds a complementary perspective by emphasizing the frictions in temporal, social and organizational that limit individuals ability to convert potential mobility into actual practice. Here the focus shifts from capacity to constraint, revealing how life rhythms, household coordination, and institutional obligations shape mobility in ways that escape classical models.

Meanwhile, Jain and Lyons concept of “gift of travel time” [10] challenges the assumption, embedded in most transport models, that travel time is inherently wasted time. They show how travelers reinterpret and appropriate travel time for socially or personally meaningful activities. These notions are further developed by Jonas de Vos & al [5] and the work of Yannick Conernet & Al [4] among others.

This insight destabilizes the central normative aim of transport planning of time minimization and introduces a crucial distinction between perceived and measured time.

These contributions do not constitute a single unified critique. Kaufmann talk about potential and try to model the mobility by accounting for human traits; Doherty integrate systemic constraint; Jain and Lyons challenge the meaning of time and Sheller & Urry foreground the socio-material entanglements of mobility systems. Together they expose the limits of reductionist planning paradigms, but they do so along different conceptual axes. These approaches reveals that mobility cannot be captured solely through technical metrics. It is an emergent property of social practices and subjective experience.

Despite the richness of these sociological insights, their penetration into actual planning practice remains limited. Planning institutions, including those in Switzerland, still operate largely within a physical paradigm centered on infrastructure provision, network performance, and transport efficiency. While notions such as comfort, accessibility, ease of use, or safety occasionally enter planning discourse, they are not articulated within the conceptual language of motility or the lived experience of travel. This gap between theory and institutional practice motivates the present research.

2.2 The Concept of Motility

Motility is defined as the capacity of an individual or a group to be mobile in social and geographic space [11]. It is not the act of moving itself, but the potential for movement. This potential is structured by three interdependent elements:

- **Access:** This refers to the available mobility tools and opportunities within a specific context. It includes the physical transport infrastructure (e.g., roads, cycle paths, rail networks), the available services (e.g., timetables, pricing schemes, owning a vehicle), and the legal and institutional frameworks (e.g., driving licenses, residency permits). Access is the objective component of the system.
- **Skills:** These are the individual competencies required to utilize access. They include not only physical skills (e.g., the ability to cycle or drive) but also cognitive skills (e.g., reading a network map, use a ticket machine) and organizational skills (e.g., time management, trip optimization).
- **Cognitive Appropriation:** This is the most subjective and crucial element. It encompasses how individuals perceive, evaluate, and assign meaning to the access and skills available to them. It includes personal preferences, habits, values (e.g., environmental consciousness), perceptions of safety, and the social meanings attached to different modes of transport (e.g., the status symbol of a car). To give a practical example, someone might prefer taking the train due to being able to re-appropriate the travel time by socializing with friends or doing administrative tasks.

Motility is, therefore, the product of the interaction between an individual's skills, the cognitive appropriation and the individual access.

2.3 Perceptions and Experiences of Time

The perception and experience of time is not merely functions of clock time. It's shaped by a rich interplay between expectations, equipment, context, and the activities travellers undertake before, during, and between trip stages. Two elements are particularly influential: Anticipation (i) and Equipement (ii).

Whether travellers anticipate (i) the delay or duration they later encounter, and (ii) whether they are equipped that is technologically, physically, or cognitively, to use these time windows in a meaningful way. These mechanisms also reinforce one another: expectations influence whether travellers prepare themselves with equipment, while equipment can in turn reshape the experience of waiting or travelling.

- **Equipement:** Equipped time refers to the condition in which travellers have access to tools, amenities, or embodied capacities that allow them to meaningfully occupy or transform time. Such equipment ranges from passive environmental features like benches, shelters, or a television in a waiting room to personal technologies enabling entertainment, communication, or productive activities. Engaging in entertainment (watching videos, listening to music, reading), social interaction (phone calls, messaging), or productive tasks (working, studying) often alters perceived travel duration and subjective satisfaction. Even activities not reliant on material equipment like resting, sleeping, or gazing out of the window, can substantially reframe the experience of time. [4].
- **Anticipation:** As mentioned earlier, waiting is experienced differently depending on whether it is expected. Real-time information, service predictability, and prior knowledge about delays significantly shape perceived duration and emotional response. Travel satisfaction is strongly linked to deviations between expected and perceived waiting time [5]: when waiting exceeds expectations, uncertainty and frustration rise, a pattern well documented among delayed airline passengers.

Experience of time is, therefore, the product between equipment and anticipation.

2.4 Accessibility and the Institutional Perspective on Mobility

Accessibility represents the degree to which places, services, and opportunities can be reached within a given spatial and temporal context. Within the motility framework, accessibility corresponds to the *access* dimension: the tangible opportunities for movement available to an individual. It determines what movements are possible in principle, while skills and cognitive appropriation determine which of these possibilities become actual mobility practices.

In everyday discourse and most planning documents, however, mobility tends to be reduced to this accessibility dimension. When policymakers or engineers discuss “improving mobility” or “qualité de la desserte” they often mean improving accessibility: reducing travel times, increasing network coverage, or enhancing service frequency and sometime improving access for person with disabilities. This reduction overlooks the skills, perceptions, and motivation that mediate how people actually use mobility systems.

As a result, there can be a gap between the provision of infrastructure and its effective use. From an institutional perspective, accessibility is usually assessed through measurable indicators. These include physical and temporal metrics such as travel time, cost, and service frequency

(“qualité de la desserte”), but also considerations (since quite recently) of inclusivity such as the accessibility of transport for people with disabilities, affordability, and multimodal connectivity. These factors together define the “objective” structure of mobility opportunities in a given territory.

Yet, while accessibility can describe how well-connected a place is in physical terms, it cannot alone account for how people interpret and appropriate those connections. For example, two neighborhoods might have identical public transport coverage, but their residents may display very different travel behaviors due to differences in skills, habits, or perceptions of comfort and safety. In this sense, accessibility forms the structural foundation of motility.

As a mathematician would describe it : necessary but not sufficient for mobility to occur. Understanding this distinction is key to developing human-centered mobility policies that engage with how people experience, value, and enact movement in their daily lives and thus maximize the effectiveness of the measure taken.

2.5 Synthesis: Implications for Spatial Planning

This conceptual framework provides the lens through which Swiss planning documents will be analyzed. The key question is whether these documents operate with a simplistic model of mobility as movement, focusing primarily on access (e.g., building infrastructure) and objective travel time. Or, do they engage with the more complex sociological model, which would involve considering how policies affect skills and cognitive appropriation, and acknowledge the role of perceived time in shaping modal choice? The presence or absence of this more nuanced vocabulary and rationale will be a key indicator of the human-centricity of planning policy.

3 Methodology

3.1 Corpus construction

The corpus analyzed in this study consists of official cantonal master plans (*Richtpläne / plans directeurs cantonaux*) collected from the websites of all Swiss cantons. When digital versions were unavailable, cantonal offices were contacted directly. Documents available only in paper format were excluded due to logistical constraints. The corpus covers French and German documents; Italian documents from Canton Ticino were excluded as they cannot be meaningfully compared with other regions in the same language. A detailed list of documents is provided in Table 3.2, and all PDFs are available in the public repository [15].

In the early stages of the research, the corpus was analyzed using **TXM** [19], an open-source textometric platform designed for large-scale linguistic analysis. TXM was used for exploratory analysis, lemmatization, keyword frequency measurement, and correspondence analysis (AFC). However, due to limitation in filtering the corpus for the AFC and in order to improve the reproducibility and automation of the workflow, the pipeline was later reimplemented in Python. The final analysis therefore almost no longer depends on TXM with the exception of the lemmatization of the dataset, although TXM played an important role in shaping the initial methodological design.

3.2 Text extraction and preprocessing

Text was extracted from PDF documents using simple command-line tools and then annotated with metadata including canton, year, and language. All text was lemmatized using **TreeTagger** [18] and stored in XML format for structured processing.

The corpus was subsequently parsed in Python using `lxml` to build a structured token dataset containing lemma, part-of-speech (POS), and document-level metadata. POS statistics were computed to assess coverage and the proportion of unknown tokens, providing an overview of data quality.

3.3 dataset filtering, Token-level scoring, smoothing, and segmentation

Each token was assigned a score based on a curated whitelist mapping lemmas to numeric values, capturing their relevance to mobility and time concepts and mobility as a whole. This whitelist was made using TXM and the cooccurrence tool to refine iteratively and find all the words that are related to our subject. To reduce noise and emphasize regions of concentrated conceptual relevance, scores were smoothed using a Gaussian kernel. High-scoring regions were then segmented based on a quantile threshold, producing subcorpora enriched in relevant lexical content. For each segment, the corresponding lemmas were exported in readable format to facilitate qualitative inspection.

3.4 Wordlist-based keyword filtering

Four custom wordlists were created for each language: one corresponding to terms related to **motility** and **accessibility** and another to terms referring to **perceived time** and **physical time**. These lists were manually assembled and iteratively refined based on corpus exploration and where made both in french and german. Both single-word and multi-word expressions were included.

One could argue that this approach is influenced by the researcher, and would be correct for individual word. But since we have a word list, the averaging get closer to the wanted concept. This is similar to psychometric test where one item in specific might not be able to predict meaningfully the presence of a disease like depression but ten items have a much stronger probability of measuring the latent concept.

3.5 Document-level counting and matrix construction

Within the extracted segments, occurrences of wordlist entries were counted per document. Counts were organized into matrices, with rows representing lemmas and columns representing documents. Separate matrices were constructed for each wordlist and aggregated into two main concept matrices: **Time** (perceived + physical) and **Mobility** (accessibility + motility). This was done for both French and German subcorpora.

3.6 Statistical analysis

Principal Component Analysis (PCA) was performed on each matrix using `scikit-learn`. PCA coordinates were normalized to a $[-1, 1]$ range for both documents and words. Visualizations display documents in blue and the top-loading lemmas in red. The explained variance of the first two components is reported to assess the dimensionality captured by the PCA. PCA was performed separately for French and German, as well as on merged bilingual matrices.

3.7 Reproducibility and automation

The entire workflow is implemented in Python and fully reproducible using open-source libraries: `lxml`, `pandas`, `numpy`, `matplotlib`, `scikit-learn`, `tqdm`, and `adjustText`. From XML parsing to KWIC generation, scoring, smoothing, segmentation, and PCA, all steps are automated in a single script (`analysis.py`). All intermediate and final outputs (counts sheet, segment texts, matrices and PCA plots) are saved in the `result/` folder to facilitate verification and replication.

Canton	Version	Year	Source	Language
AG	2	2025	Online	DE
	1	1996	†	DE
AI	4	2017	Online	DE
	3	2015	Online	DE
	2	2011	Online	DE
	1	2002	Online	DE
AR	5	2025	Online	DE
	4	2012	†	DE
	3	2008	†	DE
	2	2002	†	DE
	1	1987	†	DE
BE	10	2024	Online	FR
	10	2024	Online	DE
	8	2013	†	DE
	7	2012	†	DE
	6	2010	†	DE
	5	2008	†	DE
	4	2006	†	DE
	3	2004	†	DE
	2	2002	†	DE
	9	2016	†	FR
	8	2013	†	FR
	7	2012	†	FR
	6	2010	†	FR
	5	2008	†	FR
	4	2006	†	FR
	2	2002	†	FR
	1	1987	†	?
BL	2	2025	Online	DE
	1		†	DE
BS	7	2025	Online	DE
	6	2024	†	DE
	5	2019	†	DE
	4	2018	†	DE
	3	2012	†	DE
	2	2010	†	DE
FR	1	1986	†	DE
	3	2024	Online	FR
	2	2002	†	FR
	1	1987	†	FR
GE		2024	Online	FR
GL		2025	Online	DE
GR		2025	Online	DE
JU		2018	Online	FR

Canton	Version	Year	Source	Language
LU	4	2025	Online	DE
	3	2009	Online	DE
	2	1998	†	DE
	1	1986	†	DE
NE		2011	Online	FR
NW	4	2002	Online	DE
	3	2004	†	DE
	2	2000	†	DE
	1	1986	†	DE
OW	3	2019	Online	DE
	2	2006	†	DE
	1	1987	†	DE
SG		2017	Online	DE
SH	3	2021	Online	DE
	2	2004	†	DE
	1	198X	†	DE
SZ	6	2022	Online	DE
	5	2012	†	DE
	4	2010	†	DE
	3	2008	†	DE
	2	2004	†	DE
	1	1986	†	DE
SO	3	2023	Online	DE
	2	2000	†	DE
	1	1984	†	DE
TI	2	2009	Online	IT
	1	1990	†	IT
TG		2018	Online	DE
UR		2023	Online	DE
VD	8	2022	Online	FR
	7	2018	Online	FR
	6	2016	Online	FR
	5	2013	†	FR
	4	2012	†	FR
	3	2011	†	FR
	2	2008	†	FR
	1	198X	†	FR
VS		2019	Online	FR
ZG	3	2025	Online	DE
	2	2004	Online	DE
	1	1987	Online	DE
ZH		2020	Online	DE

Table 3.2: Cantonal master plans

Legend:

† Only available in paper format (physical consultation) and thus excluded from this study

‡ Sent upon request by the cantonal service.

* The canton with only one entry and no Version number did not answer to the email or the document whereabouts are unknown.

4 Results

4.1 Word Occurrences

In the first table 4.1 The results reveal a strong asymmetry between the two conceptual fields. Terms related to motility are almost absent from both corpora, while accessibility vocabulary is frequent and well represented. This suggests that Swiss spatial planning discourse focuses primarily on infrastructural and material dimensions of mobility. The human, cognitive, and motivational aspects that define motility are not expressed in the language of planning documents. Mobility appears as something provided through networks and infrastructure rather than as a capability that individuals develop and use. The conclusion is the same for French and German.

The same pattern is visible in the analysis of time-related terms in table 4.2. References to physical or objective time largely outweigh those to perceived or subjective time. Time is described as a variable to be reduced, optimized, or measured. Few terms evoke comfort or satisfaction. This indicates that time is approached mainly as an operational parameter rather than a human experience. But the contrast is not as dramatic as for motility between the two perspective.

Taken together, these findings show that planning discourse remains centered on measurable, supply-side factors. The subjective dimensions of mobility, including skills, motivation, and perception, are weakly represented not to say absent. The language used in official documents thus reflects and reinforces a narrow view of mobility, oriented toward infrastructure and performance rather than individual capacities and experiences.

Table 4.1: Frequency of mobility-related terms in French and German corpora (aligned by conceptual field: Motility vs Accessibility)

Category	FR term	FR hits	DE term	DE hits	Total
Motility					
	aptitude mobilité	3	Bewegungsfähigkeit	3	6
	capital mobilité	0	Mobilitätskapital	0	0
	habitus mobilité	0	Mobilitätshabitus	0	0
	autonomie déplacement	0	Bewegungsautonomie	0	0
	apprentissage mobilité	0	Mobilitätslernen	0	0
	stratégie déplacement	1	Mobilitätsstrategie	14	15
	compétence	159	Orientierungskompetenz	0	159
	représentation espace	0	Raumvorstellung	0	0
	motivation mobilité	0	Mobilitätsmotivation	0	0
	(valeur) symbolique mobilité	0	Symbolisch Mobilität	0	0
Subtotal (Motility)		163		17	180
Accessibility					
	accessibilité	284	Erreichbarkeit	207	491
	qualité desserte	159	Versorgungsqualität	3	162
	intégration transport	3	Verkehrsintegration	0	3
	coût	304	Kosten	321	625
	disponibilité transport	1	Verfügbarkeit verkehrsmittel	1	2
	infrastructure	1109	Infrastruktur	738	1847
	centralité spatial	0	räumlich zentralität	0	0
	inégalité accès	0	erreichbarkeitsgerechtigkeit	0	0
	budget mobilité	0	mobilitätsbudget	0	0
	réseau transport	331	verkehrsnetz	100	431
Subtotal (Accessibility)		2187		1370	3557

Table 4.2: Frequency of time-related terms in French and German corpora (aligned by concept)

Category	FR term	FR hits	DE term	DE hits	Total
Physical / Objective Time					
	Minute	132	Minute	207	339
	Heure	192	Stunden	0	192
	Durée	48	Dauer	2	50
	Vitesse	158	Geschwindigkeit	17	175
	Rapidité	17	Schnelligkeit	0	17
	Ponctualité	7	Pünktlichkeit	7	14
	Retard	4	Verspätung	1	5
	Kilomètre	180	Kilometer	28	208
	Ralentissement	3	Verlangsamung	3	6
	Cadence	160	Taktfrequenz	0	160
Subtotal (Physical)		901		265	1166
Perceived / Subjective Time					
	Attente	28	Wartezeit	3	31
	Stress	0	Fahrstress	0	0
	Confort	15	Komfort	0	15
	Expérience	20	Erfahrung	20	40
	Ressentir / Percevoir	5	Wahrnehmung	5	10
	agréable	10	angenehm	2	12
	Satisfaction usager	0	Fahrgastzufriedenheit	0	0
	Temps Réappropriable	0	Zeitnutzung	0	0
	temps valorisé	0	bewertete Zeit	0	0
	Temps vécu	0	gelebte Zeit	0	0
	Bien-être (mobilité)	0	Wohlfühlmobilität	0	0
Subtotal (Subjective)		78		30	108

4.2 Analysis of PCA Results related to time

As shown in Figure 4.1, the French corpus reveals a marked separation between objective and subjective temporal notions. Terms associated with measurable, operational time such as *vitesse*, *minute* and *kilomètre* are strongly loaded on the first principal component, highlighting their dominant role in structuring the discourse. In contrast, experiential notions like "stress", "confort", "attente" or "ressenti" cluster closely together and are much less present as seen in table 4.2. The cantonal points occupy a dense region in the lower-left quadrant, suggesting a relatively homogeneous and technically oriented use of time-related vocabulary across French-speaking institutional texts, with the exception of Bern, which leans more toward "cadence", "durée" and "vitesse" instead of "heure", "minute" or "kilomètre". The author theorizes that this may be due partly to a translation from German for the Bern plan, which is the only one available in both languages. The rest of the difference by the fact that Bern is also different from other german canton. Regarding Vaud, the difference can be explained by the canton hosting large commuter traffic around the Léman and dense city like Lausanne which might explain the motivation to rationalize transport more than others canton's.

Figure 4.2 shows that the German corpus follows the same pattern, but with an even stronger emphasis on objective temporal measures. Terms such as "Geschwindigkeit" and "Kilometer" form the major axis of variation and clearly structure the first component. Subjective notions like "Erfahrung" or "angenehm" rarely appear and cluster together, suggesting that they contribute weakly to the overall variance. The clustering of cantonal abbreviations in a compressed zone indicates limited lexical differentiation among German-speaking institutional texts, reinforcing the observation that technical and operational understandings of time dominate the discourse,

while experiential dimensions remain secondary with the main fracture being the usage of time versus distance. The author failed to theorize what could explain the difference Bern, AR, AI and BS between the rest of the cantons.

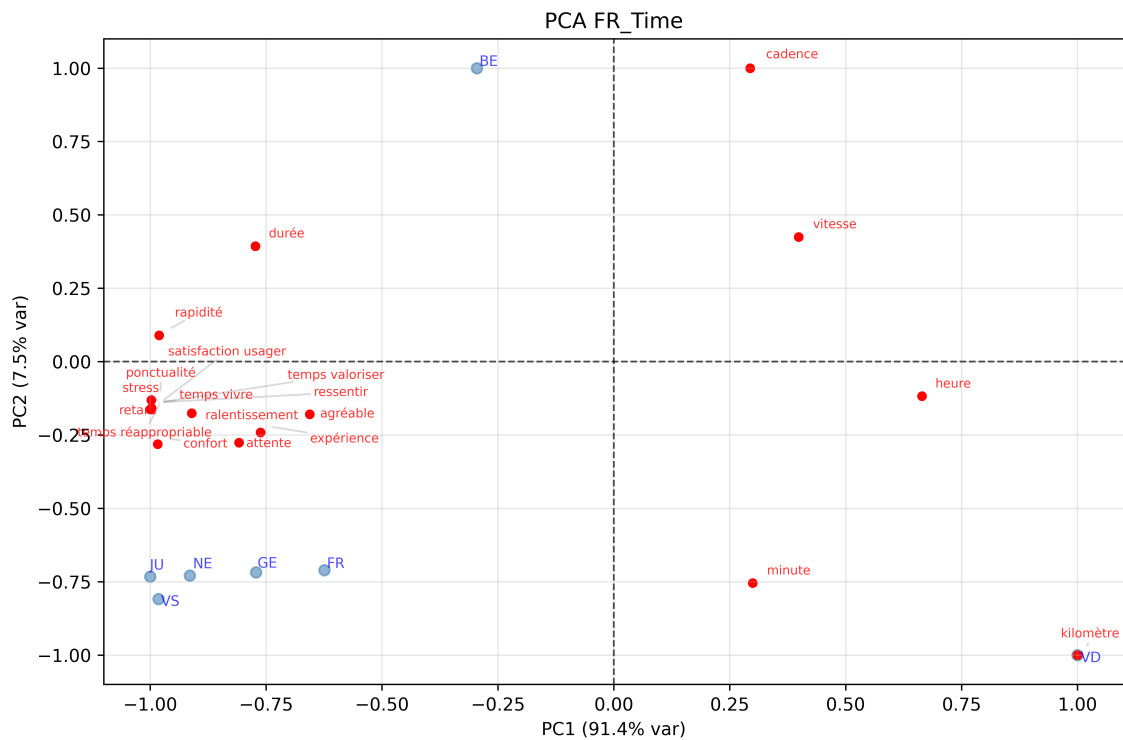


Figure 4.1: The French PCA highlights the dominance of quantifiable temporal notions, while subjective or experiential terms remain marginal in their contribution to variance.

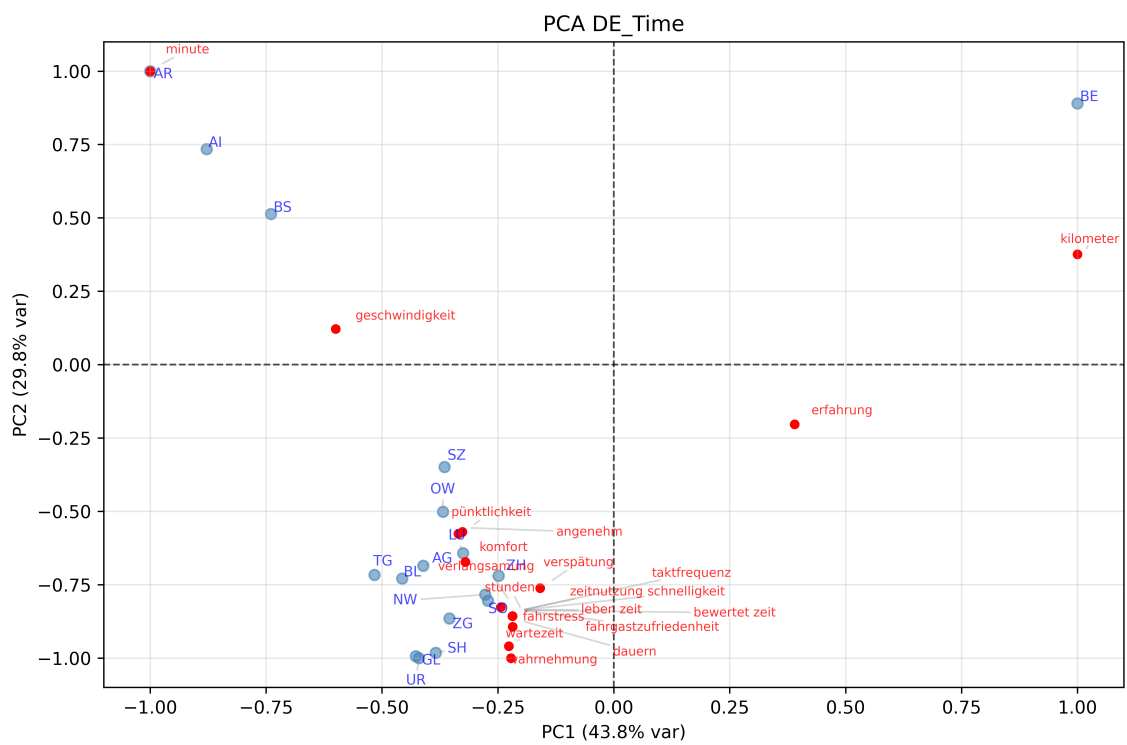


Figure 4.2: The German PCA confirms the strong prevalence of measurable time markers, with experiential notions appearing peripheral and weakly structuring the corpus.

4.3 Analysis of PCA Results related to mobility

Figure 4.3 shows a clear opposition between highly operational mobility notions and more capability-oriented terms. The first principal component captures the strong weight of "infrastructure", which dominates the positive side of the axis and forms the main structuring force in the corpus along with "coût", "accessibilité", "réseau transport" and "qualité desserte". Experiential or capability-related notions like "motivation mobilité", "apprentissage mobilité" or "habitus mobilité" cluster on the left very tightly together, indicating that they co-vary as a group but contribute only weakly to structuring differences between documents. The cantonal documents form a dense cluster near the negative side of the horizontal axis, except for Vaud and Bern, which are isolated at the positive extreme, similarly to the time PCA. This position suggests that VD relies more heavily on infrastructure and technical vocabulary than other French-speaking cantons, which use capability-based terms even more rarely than the others.

As shown in Figure 4.4, the German corpus displays a similar structure, with "infrastruktur" and "kosten" anchoring opposite sides of the second principal component. Capability-related terms such as "mobilitätskompetenz", "bewegungsautonomie" or "mobilitätshabitus" cluster near the centre and form a dense semantic block. Their central position indicates that they vary little across the documents and are very underrepresented per table 4.1; they are not the primary drivers of separation. The cantons again cluster tightly on the negative side of the horizontal axis, with BS, BL and BE positioned at the two extremes of the first two components. These outliers reflect a stronger emphasis on specific operational themes in these cantons, while the rest of the German-speaking corpus remains lexically homogeneous. Overall, measurable and infrastructural notions structure the corpus far more strongly than capability-related vocabulary.

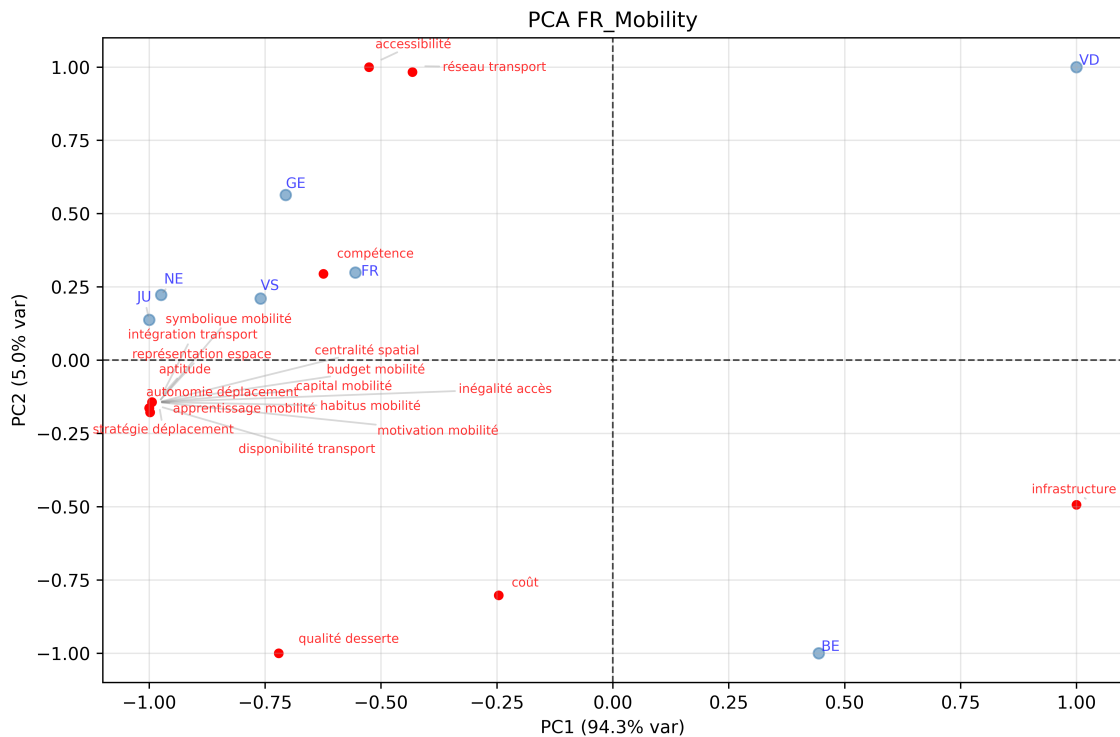


Figure 4.3: The French PCA reveals a strong axis contrasting infrastructure-focused terms with capability-oriented mobility notions, with VD standing apart from the other cantons.

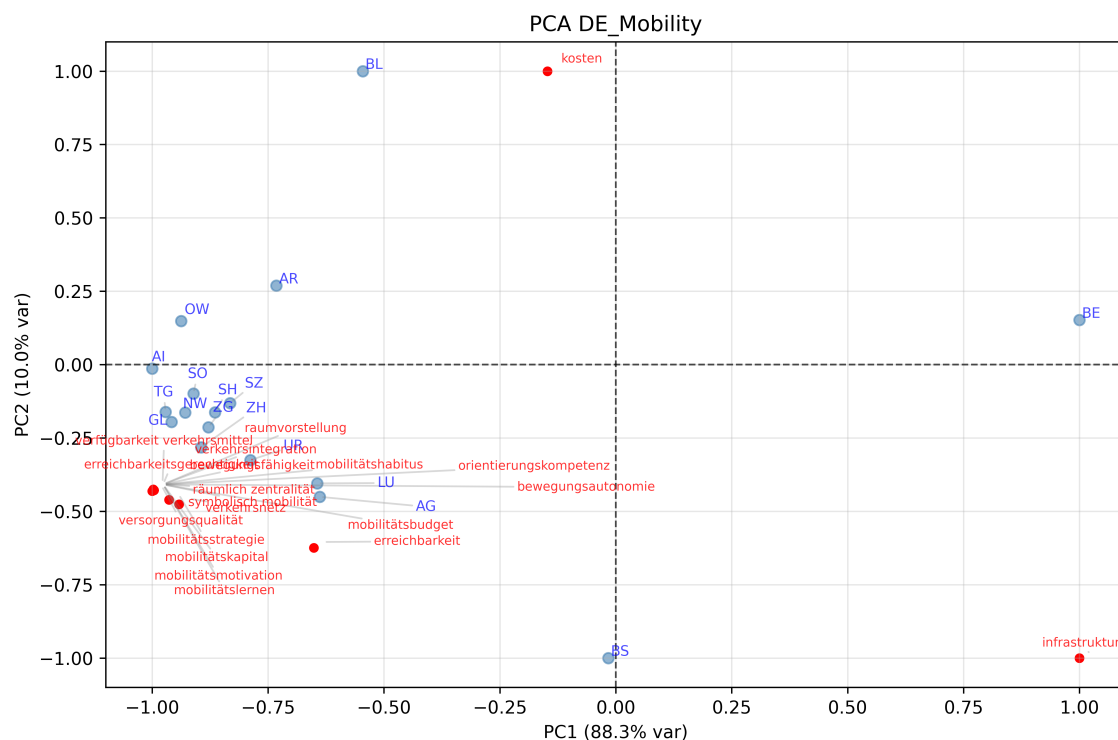


Figure 4.4: The German PCA highlights the dominance of operational and infrastructural mobility terms, while capability-related notions remain central and weakly discriminant.

4.4 Synthesis

Combining the AFC results with the word occurrence analysis reveals a coherent narrative: Swiss spatial planning discourse is heavily skewed toward measurable, supply-oriented dimensions of mobility. Accessibility and physical time terms dominate both corpora, while motility and subjective time concepts are almost absent. This is consistent across French and German texts.

Motility is conceptually underrepresented, indicating that planners primarily consider mobility as a function of infrastructure, services, and network performance rather than individual capabilities, skills or motivation. Time is similarly treated as a quantifiable variable, with minimal attention to comfort, perception or human experience.

The combined word frequency and AFC results reinforce the conclusion that Swiss planning documents operationalize mobility and time in terms of provision and efficiency. Human-centered aspects, including cognitive, motivational and experiential dimensions, remain peripheral, reflecting a narrow and infrastructure-focused view of mobility.

5 Discussion

5.1 Limitations of the Keyword Approach

The first limitation of this study concerns the use of keyword frequency as a proxy for conceptual presence. The textometric approach offers a reproducible way to measure linguistic patterns but counting words alone cannot ensure that they refer to the same idea across contexts. A high frequency of a term does not always mean that the underlying concept is strongly represented.

Future work should attempt at addressing this issue. This would strengthen the link between observed data and theoretical concepts. More advanced computational methods, such as contextual embeddings or supervised models, could help by taking into account how meaning shifts across contexts.

5.2 Challenges in Modeling Motility

A second limitation relates to how motility is modeled. Currently, there is no shared quantitative model of motility. The framework proposed by Kaufmann and later authors remains mainly qualitative, with the exception of accessibility, focused on the interaction between access, skills, and cognitive appropriation. Turning these elements into measurable indicators is a challenge.

Even if a complete model existed, combining the different dimensions into one global indicator would be problematic. From a philosophical point of view, there is no neutral way to collapse complex human phenomena into a single number. Each weighting reflects implicit values. For that reason, a multi-criteria approach such as Pareto fronts may be viable. It identifies solutions that are not dominated by others without imposing a single hierarchy. But it can only help to weed out the solution that are equal or inferior on every metrics. Since most solutions are tradeoff, they tend to be better in one metrics at the expense of another which Pareto front cannot solve. The final choice then remains with human decision-makers, who must accept the responsibility and bias that come with judgment.

5.3 Implications for Swiss Spatial Planning

The results suggest that Swiss spatial planning discourse still gives more attention to accessibility in its physical and infrastructural sense than to the human factors of mobility. Cantonal master plans mostly focus on network coverage, travel time, and service quality. They rarely address the motivation or skills that allow people to use these systems effectively. This shows that planning practice remains centered on supply-side measures, while the demand-side, related to behavior and perception, is often neglected.

Integrating motility into planning would mean designing measures that develop not only access but also the ability and willingness to use sustainable transport. This could involve highlighting the personal benefits of public transport and active mobility, and creating conditions that make their use natural and rewarding in everyday life.

5.4 Addressing the under-representation of Motivation, Skills and Experiential time

The under-representation of the “motivation” and “skills” dimensions of motility can be partly mitigated through targeted interventions. Measures such as training programs for using public or active transport, and introductory mobility workshops can help build both competence and confidence and help shape new habits. free or reduced-cost access to public transport could also help by both helping the accessibility of public transport and the motivations of the users. Interventions during key life stages, for example in schools, could encourage lasting habits of sustainable travel.

Existing research in “mobility management” [1] [16] shows that soft incentive to limit car use exist but their effect is debated [21]. Kaufmann’s approach could help improve the effectiveness of such measure by addressing the reason why people find that car are a better tradeoff from their perspective, and thus offer more effective measure that feel less frustrating to the user. This would also be complemented by taking the experiential side of time and the human experience of travel into account and thus reinforced the motivation factor.

Further work is needed to understand how these ideas could be applied systematically within Swiss planning. Operationalizing motility as a concrete planning objective remains an open challenge, both theoretical and practical.

6 Conclusion

This study examined how accessibility, motility, and time are represented in Swiss spatial planning discourse using a textometric approach. The analysis reveals a clear imbalance in planning language. Accessibility is widely present and discussed in concrete terms, whereas motility, which encompasses skills, motivation, and cognitive appropriation, is almost entirely absent. This absence indicates that capability-based dimensions of mobility are not yet part of the institutional vocabulary, making them difficult to plan for or evaluate.

Time is represented asymmetrically as well. References to physical and objective time appear frequently and are treated as operational variables to be measured or minimized. Perceived or subjective time appears to a lesser extent. Concepts such as comfort are mentioned but remain vague and underdeveloped. This suggests that planners have begun to acknowledge experiential aspects of travel, yet they do not articulate them in operational terms.

Methodologically, the textometric pipeline proved effective at revealing patterns of presence and absence in institutional language. It is valuable for mapping what planning discourse renders thinkable and actionable. At the same time, the approach is essentially descriptive. Combining it with contextual and semantic analyses would better recover meaning beyond raw counts, reducing reliance on manual interpretation and mitigating researcher bias.

The project's contributions are twofold. First, it documents a persistent discursive bias in official planning documents toward supply-side and technical framings of mobility. Second, it provides a reproducible dataset and analysis pipeline, released under a CC-BY license in the project repository [15], enabling further research or replication. Making this corpus freely available lowers the "cost" of entry and allows exploration of broader aspects of spatial planning beyond mobility alone.

Looking forward, bridging the gap between accessibility and motility requires both conceptual and institutional change. We need ways to measure motility without oversimplifying it, and planning should routinely collect data on experiential and behavioral dimensions without falling into's Goodhart's law. Only then can mobility policy move beyond networks to truly support people's capabilities and maybe let the author sleep a little easier knowing that Switzerland point a bit more toward a sustainable future.

7 Appendix

7.1 logbook

2025-08-27 kickoff meeting

2025-09-13 Mooc textometrie, recherche des sources de document

2025-09-15 Investigation de la structure entre les documents, Formation du Dataset

2025-09-25 Creation of this report, rough layout of the idea in chapter + meeting with supervisor

2025-10-02 Formation recherche Zotero

2025-10-05 Définition du champ l'exical en rapport aux questions de recherche, Analysis in TXM, searching for patterns

2025-10-15 Presentation mis semestre kaufmann + Jules

2025-10-16 Script python, run against word list

2025-10-16 Made motilité word list

2025-10-22 Made german list + analysis, sent email for previous version

2025-10-24 Writing

2025-10-26 Sorting PDF + AFC setup using TXM (failure)

2025-10-27 Writing, re-run analysis with new document

2025-10-28 finished new script to make clean AFC in python

2025-10-29 added one more document received by email, writing methodology

2025-11-07 Visio avec Thomas Bühler

2025-11-10 rework AFC + wordlist

2025-11-17 analysis, result writing

2025-11-26 result writing

2025-11-27 Slide making

7.2 scripts

[Listing 1:](#) Extract text content from PDFs by language

```
mkdir -p ../TXT && for f in *.pdf; do lang="${f:0:2}";  
mkdir -p "../TXT/$lang";  
pdftotext "$f" "../TXT/$lang/${f%.pdf}.txt"; done
```

References

- [1] The Impact Of Car-free Housing Districts On Mobility Behaviour - Case Study. URL: <https://www.witpress.com/elibrary/wit-transactions-on-ecology-and-the-environment/67/1979>.
- [2] RS 741.41 - Ordonnance du 19 juin 1995 concernan... | Fedlex. URL: https://www.fedlex.admin.ch/eli/cc/1995/4425_4425_4425/fr.
- [3] CO2 emissions from cars: facts and figures (infographics), March 2019. URL: <https://www.europarl.europa.eu/topics/en/article/20190313STO31218/co2-emissions-from-cars-facts-and-figures-infographics>.
- [4] Worthwhile travel time: a conceptual framework of the perceived value of enjoyment, productivity and fitness while travelling. *Transport Reviews*, 42(5):580–603, September 2022. URL: <https://www.sciencedirect.com/org/science/article/pii/S2324993522006303>, doi:10.1080/01441647.2021.1983067.
- [5] Jonas De Vos, Alireza Ermagun, and F. Atiyya Shaw. Wait time, travel time and waiting during travel: existing research and future directions. *Transport Reviews*, 43(5):805–810, September 2023. Publisher: Routledge. URL: <https://www.tandfonline.com/doi/full/10.1080/01441647.2023.2220206>, doi:10.1080/01441647.2023.2220206.
- [6] Catherine Doherty. Agentive Motility Meets Structural Viscosity: Australian Families Relocating in Educational Markets. *Mobilities*, 10(2):249–266, March 2015. Publisher: Routledge _eprint: <https://doi.org/10.1080/17450101.2013.853951>. doi:10.1080/17450101.2013.853951.
- [7] Assemblée fédérale de la Confédération suisse. RO 2014 899, 2014. URL: <https://www.fedlex.admin.ch/filestore/fedlex.data.admin.ch/eli/oc/2014/197/fr/pdf-a/fedlex-data-admin-ch-eli-oc-2014-197-fr-pdf-a.pdf>.
- [8] Karst T. Geurs and Bert van Wee. Accessibility evaluation of land-use and transport strategies: review and research directions. *Journal of Transport Geography*, 12(2):127–140, June 2004. URL: <https://www.sciencedirect.com/science/article/pii/S0966692303000607>, doi:10.1016/j.jtrangeo.2003.10.005.
- [9] Feng Guo, Jianhe Du, Kazuyuki Neki, Lama Bou Mjahed, Georges Bianco Darido, Wei Wang, and Ana Waksberg Guerrini. The Case for Cycling Infrastructure Investments.
- [10] Juliet Jain and Glenn Lyons. The gift of travel time. *Journal of Transport Geography*, 16(2):81–89, March 2008. URL: <https://www.sciencedirect.com/science/article/pii/S0966692307000531>, doi:10.1016/j.jtrangeo.2007.05.001.
- [11] Vincent Kaufmann, Manfred Max Bergman, and Dominique Joye. Motility: mobility as capital. *International Journal of Urban and Regional Research*, 28(4):745–756, 2004. _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.0309-1317.2004.00549.x>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.0309-1317.2004.00549.x>, doi:10.1111/j.0309-1317.2004.00549.x.
- [12] Todd Litman. Evaluating Transportation Equity: Guidance for Incorporating Distributional Impacts in Transportation Planning. *Social Research in Transport (SORT) Clearinghouse*, 8, January 2007.
- [13] Antoine Maier, Aude Maier, and Tom David. Take Goodhart Seriously: Principled Limit on General-Purpose AI Optimization, October 2025. arXiv:2510.02840 [cs]. URL: <http://arxiv.org/abs/2510.02840>, doi:10.48550/arXiv.2510.02840.

- [14] Robert K. Merton. The Unanticipated Consequences of Purposive Social Action. *American Sociological Review*, 1(6):894–904, 1936. Publisher: [American Sociological Association, Sage Publications, Inc.]. URL: <https://www.jstor.org/stable/2084615>, doi:10.2307/2084615.
- [15] Nathann Morand. nathmo/SwissSpatialPlanningTextometricAnalysis, November 2025. original-date: 2025-10-16T14:07:40Z. URL: <https://github.com/nathmo/SwissSpatialPlanningTextometricAnalysis>.
- [16] Guido Möser and Sebastian Bamberg. The effectiveness of soft transport policy measures: A critical assessment and meta-analysis of empirical evidence. *Journal of Environmental Psychology*, 28(1):10–26, March 2008. URL: <https://www.sciencedirect.com/science/article/pii/S0272494407000722>, doi:10.1016/j.jenvp.2007.09.001.
- [17] United Nations. Universal Declaration of Human Rights. Publisher: United Nations. URL: <https://www.un.org/en/about-us/universal-declaration-of-human-rights>.
- [18] H. Schmid. Improvements in Part-of-Speech Tagging with an Application to German. In Susan Armstrong, Kenneth Church, Pierre Isabelle, Sandra Manzi, Evelyne Tzoukermann, and David Yarowsky, editors, *Natural Language Processing Using Very Large Corpora*, pages 13–25. Springer Netherlands, Dordrecht, 1999. doi:10.1007/978-94-017-2390-9_2.
- [19] Heiden Serge. The TXM Platform: Building Open-Source Textual Analysis Software Compatible with the TEI Encoding Scheme. pages 389–398, 2010. URL: <http://halshs.archives-ouvertes.fr/docs/00/54/97/64/PDF/paclic24.sheiden.pdf>.
- [20] Mimi Sheller and John Urry. The New Mobilities Paradigm. *Environment and Planning A*, 38:207–226, February 2006. doi:10.1068/a37268.
- [21] Rob Wall, Werner Brög, Erhard Erl, James Ryle, and Franz Barta. A response to: Möser, G., & Bamberg, S. (2008). The effectiveness of soft transport policy measures: A critical assessment and meta-analysis of empirical evidence. *Journal of Environmental Psychology*, 28, 10–26. *Journal of Environmental Psychology*, 31(3):266–269, September 2011. URL: <https://www.sciencedirect.com/science/article/pii/S0272494411000387>, doi:10.1016/j.jenvp.2011.05.004.